

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Project: CommercePilot

Version: 1.0

Status: Approved

Document Owner: Product Team

Project Type: AI-Powered Multi-Tenant SaaS Platform

1. Executive Summary

CommercePilot is an AI-powered commerce automation platform designed to integrate with existing e-commerce systems such as WooCommerce, Shopify (future), and custom online stores.

Instead of replacing existing platforms, CommercePilot acts as an intelligent automation layer that understands customer conversations, automates repetitive business processes, and assists business owners with AI-driven decision support while maintaining human approval over critical operations.

The primary objective of the MVP is to automate WhatsApp-based order processing and synchronize approved orders with an existing WooCommerce store.

2. Problem Statement

Many SMEs receive customer orders through WhatsApp instead of directly through their websites.

Business owners currently perform the following tasks manually:

- Reading customer messages
- Understanding customer intent
- Identifying requested products
- Checking inventory
- Creating orders manually
- Sending quotations
- Confirming payments
- Updating customers
- Maintaining order records

This results in:

- Slow response times
- Human errors

- Lost orders
 - High operational costs
 - Poor customer experience
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3. Product Goal

Build a production-ready AI automation platform capable of:

- Understanding customer messages
 - Extracting structured order information
 - Validating product availability
 - Presenting AI-generated orders for owner approval
 - Automatically creating approved orders in WooCommerce
 - Sending customer confirmations
 - Recording every action for auditing
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4. Target Users

Primary Users

- SME business owners
- Online retailers
- Textile manufacturers
- Fashion stores
- Electronics shops

Secondary Users

- Sales staff
- Customer support agents
- Warehouse operators

System Users

- Super Admin
 - Tenant Owner
 - Staff Member
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5. MVP Scope

Included

- Multi-tenant architecture
- Authentication
- WhatsApp Cloud API integration

- AI message understanding
- Product matching using RAG
- Order extraction
- Owner approval dashboard
- WooCommerce integration
- Email notifications
- Audit logs
- Activity history

Excluded

- Shopify integration
 - Facebook Messenger
 - Instagram
 - Mobile application
 - Voice order processing
 - Multi-language AI (Phase 2)
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6. Functional Requirements

Authentication

The system shall:

- Register business owners
 - Authenticate users securely
 - Support role-based access control
 - Support tenant isolation
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WhatsApp Integration

The system shall:

- Receive incoming messages
 - Store message history
 - Process text messages
 - Track delivery status
 - Queue AI processing
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AI Processing

The AI shall:

- Detect customer intent
- Extract products

- Extract quantity
 - Extract size
 - Extract colour
 - Extract delivery information
 - Estimate confidence score
 - Never invent products
 - Use RAG against the tenant's product catalogue
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Order Management

The system shall:

- Generate draft orders
 - Allow owner review
 - Approve or reject orders
 - Synchronize approved orders with WooCommerce
 - Record all changes
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Dashboard

Provide:

- Pending AI orders
 - Approved orders
 - Rejected orders
 - AI confidence
 - Customer history
 - Analytics
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Notifications

The system shall:

- Send owner email notifications
 - Send WhatsApp confirmations
 - Notify on approval/rejection
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7. Non-Functional Requirements

- Production-ready architecture
- High availability
- Secure authentication
- Modular services
- REST API

- Event-driven processing
 - Horizontal scalability
 - Comprehensive logging
 - Unit and integration tests
 - API documentation
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8. User Stories

US-001

As a customer,

I want to place an order through WhatsApp,

So that I do not need to visit the website.

US-002

As a business owner,

I want AI to summarize customer orders,

So that I can approve them quickly.

US-003

As a business owner,

I want approved orders to appear automatically in WooCommerce,

So that I do not need manual data entry.

US-004

As a staff member,

I want to see customer conversation history,

So that I understand the order context.

US-005

As an administrator,

I want complete audit logs,

So that every action is traceable.

9. Acceptance Criteria

The MVP is complete when:

- ✓ Customer sends WhatsApp message
 - ✓ AI extracts structured order
 - ✓ AI validates products using RAG
 - ✓ AI generates confidence score
 - ✓ Owner receives notification
 - ✓ Owner reviews order
 - ✓ Owner approves
 - ✓ WooCommerce order created automatically
 - ✓ Customer receives confirmation
 - ✓ Audit log recorded
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10. Business Rules

- AI cannot approve orders automatically unless explicitly enabled.
 - Products must exist in the tenant catalogue.
 - Every order belongs to one tenant.
 - Every action must be logged.
 - Business owners always have the final decision.
 - Failed integrations must be retried.
 - Sensitive customer data must be protected.
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11. Success Metrics

Business Metrics

- 80% reduction in manual order entry

- 90% reduction in order processing time
- 95% order extraction accuracy
- Customer response under 1 minute

Technical Metrics

- API response < 500 ms
 - AI processing < 10 seconds
 - System uptime > 99%
 - Zero cross-tenant data leakage
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12. Future Roadmap

Phase 2

- Shopify integration
- Facebook Messenger
- Instagram DM
- Sinhala language support
- Voice message processing

Phase 3

- Inventory prediction
 - AI sales forecasting
 - Marketing automation
 - Courier integration
 - Plugin marketplace
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13. Risks

- AI hallucinations
- Incorrect product matching
- API rate limits
- WhatsApp template restrictions
- Third-party API failures

Mitigation:

- RAG
 - Human approval
 - Retry queues
 - Audit logging
 - Monitoring
 - Confidence scoring
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14. Definition of Done

A feature is complete only if:

- Business requirements satisfied
- Production-quality code
- Unit tests written
- Integration tests pass
- Documentation updated
- Security reviewed
- Error handling implemented
- Logging implemented
- Code reviewed
- No critical bugs remain